

MSC ARTIFICIAL INTELLIGENCE
MASTER THESIS

General Purpose Geometrically Aware Hyperbolic Residual-Quantized Variational AutoEncoders

by
ALESSIO COLOMBO
15387569

June 4, 2026

36 ETCs
January 6th 2026 to June 19th 2026

Supervisor:
Melika Ayoughi

Examiner:
Dr A PERSON

Second reader:
Dr A PERSON



UNIVERSITEIT VAN AMSTERDAM

Contents

1	Introduction	1
2	Related Work	3
2.1	Vector Quantization	3
2.2	Residual Vector Quantization	3
2.3	Hyperbolic Representation Learning	4
3	Background	6
3.1	Vector-Quantized Variational Autoencoders	6
3.2	Residual Vector Quantization	7
3.3	Hyperbolic Geometry and the Poincaré Ball	7
3.3.1	The Poincaré Ball Model	8
3.3.2	Gyrovector Spaces and Möbius Operations	8
3.3.3	Geodesic Distance	8
3.3.4	Tangent Spaces and the Exponential and Logarithmic Maps	9
3.3.5	Parallel Transport	9
3.3.6	Riemannian Gradients	9
4	Methods	10
4.1	Hyperbolic Residual Quantization	10
4.1.1	Hyperbolic Distance and Quantization	10
4.1.2	The Hyperbolic Straight-Through Estimator	11
4.2	The Residual Firewall and Gradient Routing	12
4.2.1	The Euclidean firewall and its breakdown	13
4.2.2	Compounding across stacked transports	13
4.2.3	Spectral radius and the conformal amplifier	13
4.2.4	Restoring the firewall: the gradient-routing correction	14
4.3	Residual Aggregation: the <code>new_method</code> Convention	14
5	Experiments	16
6	Conclusions	17
A	Optional appendix	18

Abstract

Provide a short overview and description of your thesis here.

Chapter 1

Introduction

In recent years, representation learning has increasingly shifted toward the discretization of diverse data modalities into token sequences, which underlies the scalability of modern multimodal models. By converting continuous, high-dimensional signals (such as images, audio, and video) into sequences of discrete tokens, this approach enables the effective application of powerful generative architectures, including autoregressive transformers and diffusion models. A prominent framework for discretizing continuous data is provided by autoencoders based on Vector Quantization [?, ?, ?, ?, ?], along with their hierarchical extension, Residual Quantized VAEs (RQ-VAEs) [?]. In particular, RQ-VAEs support the coarse-to-fine decomposition of data by constructing hierarchical representations through successive stages of quantization, thereby capturing structures at multiple levels of granularity.

Despite their empirical success, conventional quantization methods have fundamental structural limitations. These methods are also prone to well-known optimization issues, including codebook collapse, vanishing residual contributions, codebook collisions, and scaling instabilities [?]. Another fundamental geometric limitation of these approaches is their reliance on Euclidean latent spaces. Euclidean geometry is intrinsically flat, with volume growing only polynomially as a function of the radius, making it poorly suited for representing hierarchical data. This mismatch becomes particularly pronounced in the context of RQ-VAEs, which naturally induce tree-like structures: early quantization stages capture coarse information, while subsequent stages progressively refine it. Such hierarchical organizations are more faithfully represented in hyperbolic space, where constant negative curvature induces exponential volume growth. This geometric property closely mirrors the expansion behavior of trees, providing a more appropriate inductive bias for hierarchical representation learning.

The exponential growth of volume induced by this metric makes hyperbolic space a natural embedding space for the hierarchical representations produced by residual quantization.

Although recent work has explored hyperbolic extensions of VQ-VAEs, a critical methodological gap remains. Existing approaches often directly transplant Euclidean training procedures—such as standard K-means clustering or exponential moving average (EMA) updates—into hyperbolic manifolds. However, such techniques fail to account for the intrinsic geometry of these spaces, including the role of Fréchet means and the non-Euclidean distance structure defined in models such as the Poincaré ball and Lorentz manifold. As a result, these methods operate in hyperbolic spaces while retaining fundamentally Euclidean optimization principles.

In this study, we introduce a novel framework for *Hyperbolic Residual Quantization*, in which the quantization process is treated as a first-class operation on a hyperbolic manifold. Instead of relying on naive geometric projections, we designed training objectives that explicitly leveraged the curvature of the space during codebook updates.

To show that our changes are generalizable, we evaluated the proposed approach across

two domains: image synthesis (for visualizations) and audio modeling (because the results of applying hyperbolic learning to audio are promising). To the best of our knowledge, this is the first application of hyperbolic RQ-VAEs to audio data, where hierarchical structures such as harmonic and temporal patterns are particularly prominent. Furthermore, we provide a detailed analysis of the learned latent representations at inference time, demonstrating that hyperbolic quantization yields improved codebook utilization and a more interpretable hierarchical organization than Euclidean baselines.

Chapter 2

Related Work

This chapter reviews the literature relevant to our work, placing our contributions in a broader context by first discussing discrete representation learning through vector quantization (§2.1), then examining its hierarchical extension through residual quantization (§2.2), and finally reviewing the application of hyperbolic geometry to representation learning (§2.3).

2.1 Vector Quantization

Vector quantization (VQ) has a long history in signal processing as a method for lossy data compression [?]. The modern revival of VQ in deep learning was catalyzed by the VQ-VAE [?], which introduced a discrete bottleneck into the variational autoencoder framework. By mapping continuous encoder outputs to the nearest vector in a learnable codebook, VQ-VAE enabled the learning of compact discrete representations suitable for downstream generative modeling. Since the nearest-neighbor assignment is non-differentiable, gradient flow through the quantization step is enabled by the Straight-Through Estimator (STE) [?], which copies gradients from the decoder input directly to the encoder output, effectively treating the quantization as the identity function during backpropagation.

To mitigate the well-known issue of codebook collapse [?, ?], various remedies have been proposed, ranging from heuristic codebook resets and exponential moving average (EMA) updates to principled variational relaxations (e.g., HQ-VAE [?]) and alternative discrete bottlenecks like Gumbel-Softmax [?] and Finite Scalar Quantization [?].

While single-stage VQ enables high-fidelity synthesis and generation, achieving high representational capacity requires exponentially large codebooks, which are difficult to optimize. Alternatively, increasing the spatial sequence length of discrete tokens creates a computational bottleneck for downstream autoregressive transformers, which scale quadratically with sequence length. These limitations motivate the multi-stage residual approach discussed next.

2.2 Residual Vector Quantization

Residual Vector Quantization (RVQ) extends the single-stage VQ paradigm by quantizing the representation in multiple successive stages [?]. At each stage d , the residual error from the previous stage is quantized using a dedicated codebook, and the new residual is computed by subtracting the selected codeword. The final representation is the sum of all selected codewords across the D stages. This coarse-to-fine decomposition effectively constructs a virtual codebook with exponential representational capacity (K^D for D codebooks of size K) without expanding the physical memory footprint or the spatial sequence length of the discrete tokens. A single input datum is thus represented not by a single discrete token, but by an ordered tuple of D

indices, where early entries capture global, macro-level information and later entries encode progressively finer details.

RQ-VAE has been successfully applied across a wide range of domains. In image generation, RVQ reduces the sequence length for autoregressive modeling while maintaining high fidelity [?], and has been further extended to discrete diffusion models [?], progressive generative image compression [?], and unified visual understanding and generation frameworks [?]. In the audio domain, foundational RVQ-based codecs like SoundStream [?] and EnCodec [?] provide variable-bitrate compression and serve as the standard discrete tokenizers for large-scale language-model-based audio and music generation systems [?, ?]. This foundational role in audio has since expanded into diverse applications such as low-latency speech synthesis [?], speech enhancement [?], multimodal generation [?], and deepfake detection [?]. Beyond perceptual tasks, RVQ has been widely adopted in generative recommender systems. Originating with frameworks like TIGER [?] that assign items hierarchical semantic identifiers for zero-shot generalization, this approach has rapidly advanced. Recent developments leverage RVQ for conversational recommendation [?], discrete diffusion-based bundle construction [?], stabilizing generation via reference vectors [?], and accommodating the power-law distribution of user-item interactions through hyperbolic spaces [?].

A consistent property across all of these applications is the inherent hierarchical structure induced by the residual decomposition: truncating the code tuple at any depth yields a progressively coarser but semantically coherent approximation of the input, enabling natively progressive data transmission and flexible rate-distortion trade-offs. However, all existing RVQ methods operate entirely in Euclidean space, where the flat geometry and polynomial volume growth are fundamentally mismatched with the tree-like hierarchical structure that residual quantization naturally induces.

2.3 Hyperbolic Representation Learning

Hyperbolic geometry offers a continuous analogue of discrete tree structures. In a d -dimensional hyperbolic space of constant negative curvature, the volume of a geodesic ball grows exponentially with its radius, in stark contrast to the polynomial growth of Euclidean space [?]. This geometric property makes hyperbolic manifolds particularly well-suited for embedding hierarchical data with low distortion. Sala et al. [?] formalized this connection by proving that any weighted tree with n nodes can be embedded into a two-dimensional hyperbolic space with arbitrarily low distortion, whereas achieving comparable fidelity in Euclidean space requires $\mathcal{O}(n)$ dimensions.

The application of hyperbolic geometry to representation learning was pioneered by Nickel and Kiela [?], who demonstrated that Poincaré embeddings efficiently capture large-scale taxonomies with far fewer dimensions. Ganea et al. [?] formalized the extension of neural network operations to hyperbolic space using Möbius gyrovector algebra [?], allowing entire feed-forward architectures to operate in the Poincaré ball. Since then, hyperbolic neural networks have been successfully applied across diverse domains, including computer vision [?, ?, ?] and audio source separation [?]. Closer to our setting, Mathieu et al. [?] introduced the Poincaré Variational Autoencoder ($\mathcal{P}$ -VAE), demonstrating that hierarchical structure naturally emerges in hyperbolic latent spaces without explicit supervision. More recently, several works have explored combining hyperbolic geometry with vector quantization. Hyperbolic VQ-VAE approaches map continuous data and discrete codebooks into hyperbolic space. Methods such as HyperVQ [?] formulate codebook assignment via hyperbolic logistic regression, while others like HVQ-VAE [?] embed both data and codebooks directly into the Poincaré ball and adapt clustering to the manifold. These hyperbolic discrete bottlenecks have seen rapid adoption across fields, including graph generation [?], semantic tokenization [?, ?], and audio processing [?]. However,

a recurring limitation across these VQ-VAE extensions is their reliance on Euclidean training machinery. Specifically, the Straight-Through Estimator (STE) copies gradients through the identity function, ignoring the fact that gradient transport on a curved manifold requires proper Riemannian operations.

Building on these single-stage quantization methods, works such as Hyperbolic Residual Quantization (HRQ) [?] and its recent applications in generative recommender systems [?] extend the multi-stage residual paradigm to the Poincaré ball. To compute hierarchical residuals, HRQ replaces standard Euclidean addition and subtraction with their Möbius counterparts. However, this introduces significant limitations in residual computation. Because Möbius addition is neither commutative nor associative, the sequential aggregation of residual codes becomes mathematically complex and prone to geometric inconsistencies and numerical instability across multiple stages. In short, while existing approaches successfully place representations on hyperbolic manifolds, they struggle to fully adapt backpropagation and residual computation to the curvature of the space. Bridging this gap—by making the fundamental operations of both quantization and residual aggregation genuinely hyperbolic—is the central objective of this work.

Chapter 3

Background

This chapter introduces the technical machinery on which the remainder of the thesis builds. We first review discrete representation learning through Vector Quantization (§3.1) and its multi-stage extension, Residual Vector Quantization (§3.2), paying particular attention to the Straight-Through Estimator and the gradient flow it induces, as these are central to the methods developed later. We then present the elements of hyperbolic geometry required to lift these operations onto a curved manifold (§3.3): the Poincaré ball model, the gyrovector formalism that endows it with algebraic structure, the exponential and logarithmic maps, parallel transport, and the relationship between Euclidean and Riemannian gradients. Throughout, we restrict the exposition to the constructs that are strictly necessary to formulate the residual aggregation scheme, the hyperbolic Straight-Through Estimator, the gradient-routing correction, and the numerical-stabilization techniques introduced in Chapter 4.

3.1 Vector-Quantized Variational Autoencoders

The Vector Quantized Variational Autoencoder (VQ-VAE) [?] learns a discrete latent representation of continuous data. Given an input signal x , an encoder network E maps it to a continuous latent representation $z_e = E(x) \in \mathbb{R}^D$. A trainable codebook $C = \{c_1, \dots, c_K\} \subset \mathbb{R}^D$ of K entries is then used to discretize this continuous vector. The quantization operator selects the nearest codeword under the Euclidean distance:

$$q(z_e) = c_k, \quad \text{where } k = \operatorname{argmin}_j \|z_e - c_j\|_2^2. \quad (3.1)$$

The selected vector $q(z_e)$ is subsequently passed to a decoder G that reconstructs the original signal as $\hat{x} = G(q(z_e))$.

The Straight-Through Estimator. The nearest-neighbour assignment argmin is a piecewise-constant function, so its derivative with respect to z_e is zero almost everywhere and undefined at cell boundaries. Consequently, no gradient can flow from the decoder back to the encoder through the quantization step. The VQ-VAE circumvents this with the Straight-Through Estimator (STE) [?], which copies the gradient from the quantized output directly to the encoder output, treating the quantizer as the identity during backpropagation. Operationally, the forward value is computed while the difference $q(z_e) - z_e$ is held under the stop-gradient operator $\operatorname{sg}[\cdot]$:

$$\hat{z}_{\text{STE}} = z_e + \operatorname{sg} [q(z_e) - z_e]. \quad (3.2)$$

This expression evaluates to $q(z_e)$ in the forward pass, yet because the stop-gradient term is treated as a constant, its Jacobian reduces to $\partial \hat{z}_{\text{STE}} / \partial z_e = I$, so the decoder gradient is passed unaltered to the encoder.

Training objective. Since the codebook is not reached by the reconstruction gradient through the STE, VQ-VAE is trained with the composite objective

$$\mathcal{L} = \underbrace{\|x - \hat{x}\|_2^2}_{\text{reconstruction}} + \underbrace{\|\text{sg}[z_e] - q(z_e)\|_2^2}_{\text{codebook}} + \beta \underbrace{\|z_e - \text{sg}[q(z_e)]\|_2^2}_{\text{commitment}}, \quad (3.3)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator and β is a weighting hyperparameter. The codebook term pulls the selected centroids towards the encoder outputs, whereas the commitment term pulls the encoder outputs towards their assigned centroids, preventing the continuous representation from diverging from the discrete dictionary. A persistent difficulty of this framework is *codebook collapse*, in which only a small fraction of entries is ever selected [?]. We mitigate it with a *dead-code revival* heuristic: the usage of each codebook entry is tracked across batches, and any entry whose usage falls below a fixed threshold τ_{dead} is flagged as dead and reinitialized to a randomly selected encoder output from the current batch. This continually recycles under-utilized codewords towards regions of the latent space that the encoder actually populates, keeping codebook utilization high.

3.2 Residual Vector Quantization

Residual Vector Quantization (RVQ) extends the single-stage paradigm by quantizing the latent representation in D successive stages, each with its own codebook [?]. Let $r_0 = z_e$ denote the encoder output. At stage i the current residual r_{i-1} is quantized to a codeword q_i , and the residual for the next stage is obtained by subtracting the selected codeword:

$$r_i = r_{i-1} - q_i. \quad (3.4)$$

The final representation is the sum of the selected codewords across all stages, $\hat{z} = \sum_{i=1}^D q_i$. This coarse-to-fine decomposition constructs a virtual codebook of effective size K^D without enlarging the physical dictionary or the token sequence length, and induces a natural hierarchy: early stages capture coarse, global structure while later stages encode progressively finer detail. This makes RVQ especially effective for multi-scale signals such as audio, where global structure is resolved in the initial stages and fine-grained acoustic texture in the deeper ones.

Gradient routing in Euclidean RVQ. The gradient flow induced by stacking STEs is a property we will rely on repeatedly, so we make it explicit here. Combining the STE Jacobian $\partial q_i / \partial r_{i-1} = I$ (Eq. 3.2) with the residual update of Eq. 3.4, the Jacobian of the residual branch is

$$\frac{\partial r_i}{\partial r_{i-1}} = \frac{\partial (r_{i-1} - q_i)}{\partial r_{i-1}} = I - I = 0. \quad (3.5)$$

The residual branch therefore transmits no gradient: each codebook is reached by the reconstruction signal directly through its own q_i , while the encoder receives exactly one clean copy of the gradient, regardless of the number of stages. This exact cancellation, which we will refer to as the residual gradient *firewall*, keeps the magnitude of the gradient delivered to the encoder stable as the depth D grows. As shown in Chapter 4, this cancellation is specific to the flat geometry and does not survive the transition to a curved manifold.

3.3 Hyperbolic Geometry and the Poincaré Ball

Hyperbolic space is a Riemannian manifold of constant negative curvature. Its defining feature is that the volume of a geodesic ball grows *exponentially* with its radius, as opposed to the

polynomial growth of Euclidean space [?]. This expansion rate matches that of the node count of a regular tree, which is why hyperbolic spaces embed hierarchical and tree-like data with far lower distortion than Euclidean spaces of comparable dimension [?, ?]. Among the several isometric models of hyperbolic geometry, we adopt the Poincaré ball, which is well suited to gradient-based learning because its operations admit closed forms and its points are easily constrained to a bounded region [?].

3.3.1 The Poincaré Ball Model

For a curvature parameter $c > 0$, the Poincaré ball is the open ball

$$\mathbb{D}_c^D = \{ x \in \mathbb{R}^D : c \|x\|^2 < 1 \}, \quad (3.6)$$

of radius $1/\sqrt{c}$, equipped with the Riemannian metric

$$g_x^c = (\lambda_x^c)^2 g^E, \quad \lambda_x^c = \frac{2}{1 - c \|x\|^2}, \quad (3.7)$$

where g^E is the Euclidean metric and λ_x^c is the *conformal factor*. The metric is conformal, meaning it rescales Euclidean lengths isotropically by λ_x^c without altering angles. The setting $c = 0$ recovers Euclidean space as a limiting case. The behaviour of λ_x^c is decisive for everything that follows: at the origin $\lambda_0^c = 2$, but as a point approaches the boundary, $c \|x\|^2 \rightarrow 1$ and $\lambda_x^c \rightarrow \infty$. Riemannian distances near the boundary are thus enormously magnified relative to Euclidean ones, which both grants the model its expressive power and makes computations involving λ_x^c numerically delicate in that regime.

3.3.2 Gyrovector Spaces and Möbius Operations

Because vector addition does not preserve membership in the ball, the Poincaré ball is instead given an algebraic structure through the gyrovector formalism [?]. The analogue of vector addition is the Möbius addition

$$x \oplus_c y = \frac{(1 + 2c \langle x, y \rangle + c \|y\|^2) x + (1 - c \|x\|^2) y}{1 + 2c \langle x, y \rangle + c^2 \|x\|^2 \|y\|^2}, \quad (3.8)$$

which maps any two points of $\mathbb{D}_c^D$ to a point inside the ball, and the corresponding Möbius subtraction $x \ominus_c y = x \oplus_c (-y)$. Crucially, and in contrast with Euclidean addition, $\oplus_c$ is neither commutative nor associative; the order in which a sequence of points is aggregated therefore matters. The failure of commutativity is captured precisely by the *gyration* operator

$$\text{gyr}[u, v] w = \ominus(u \oplus_c v) \oplus_c (u \oplus_c (v \oplus_c w)), \quad (3.9)$$

an automorphism of the ball that acts as a rotation: $u \oplus_c v = \text{gyr}[u, v] (v \oplus_c u)$. The gyration encodes the holonomy of the curved space and, as we will see, is the source of the rotational mismatch that breaks the Euclidean gradient firewall.

3.3.3 Geodesic Distance

The geodesic distance induced by the metric 3.7 admits the closed form

$$d_{\mathbb{D}_c}(x, y) = \frac{2}{\sqrt{c}} \tanh^{-1} \left(\sqrt{c} \|(-x) \oplus_c y\| \right). \quad (3.10)$$

Unlike the Euclidean metric, this distance grows without bound as either argument approaches the boundary, reflecting the exponential expansion of volume; two points near the boundary can be geodesically far apart even when their Euclidean separation is small.

3.3.4 Tangent Spaces and the Exponential and Logarithmic Maps

At every point $x \in \mathbb{D}_c^D$ the tangent space $T_x \mathbb{D}_c^D$ is a local Euclidean linearization of the manifold, which provides the natural arena for performing additive vector operations. Movement between the manifold and a tangent space is mediated by the exponential and logarithmic maps, which are mutual inverses. The exponential map $\exp_x^c : T_x \mathbb{D}_c^D \rightarrow \mathbb{D}_c^D$ moves from x along the geodesic in the direction of a tangent vector v , while the logarithmic map $\log_x^c$ is its inverse. At the origin they take the simple radial forms

$$\exp_0^c(v) = \tanh(\sqrt{c} \|v\|) \frac{v}{\sqrt{c} \|v\|}, \quad \log_0^c(y) = \tanh^{-1}(\sqrt{c} \|y\|) \frac{y}{\sqrt{c} \|y\|}, \quad (3.11)$$

and at an arbitrary base point they are obtained by combining the origin maps with a Möbius translation and the conformal factor:

$$\exp_x^c(v) = x \oplus_c \exp_0^c\left(\frac{\lambda_x^c}{2} v\right), \quad \log_x^c(y) = \frac{2}{\lambda_x^c} \log_0^c((-x) \oplus_c y). \quad (3.12)$$

These maps make it possible to carry out an operation in the flat tangent space and then return to the manifold, a pattern exploited extensively when constructing geometrically valid analogues of Euclidean procedures.

3.3.5 Parallel Transport

Tangent spaces at distinct points are different vector spaces, so a tangent vector defined at x cannot be directly compared with, or added to, one defined at y . Parallel transport $P_{x \rightarrow y}^c : T_x \mathbb{D}_c^D \rightarrow T_y \mathbb{D}_c^D$ carries a tangent vector from x to y along the connecting geodesic while preserving its Riemannian norm. On the Poincaré ball it has the closed form

$$P_{x \rightarrow y}^c(v) = \frac{\lambda_x^c}{\lambda_y^c} \text{gyr}[y, -x] v. \quad (3.13)$$

This is a genuine isometry of tangent spaces: it both rescales the vector by the ratio of conformal factors and rotates it through the gyration operator. The rotation embodied by $\text{gyr}[y, -x]$ is the essential difference between true parallel transport and a magnitude-only conformal rescaling; this distinction will prove central when transporting gradients across stages.

3.3.6 Riemannian Gradients

Because the metric 3.7 rescales the inner product by $(\lambda_x^c)^2$, the direction of steepest ascent on the manifold differs from the ordinary Euclidean gradient. For a scalar function f , the Riemannian gradient relates to the Euclidean gradient by

$$\nabla^R f(x) = \frac{1}{(\lambda_x^c)^2} \nabla^E f(x). \quad (3.14)$$

A correct backward pass on the Poincaré ball therefore involves converting Euclidean gradients to Riemannian ones (dividing by $(\lambda^c)^2$), transporting them between tangent spaces via Eq. 3.13, and converting back. As the conformal factor diverges near the boundary, these conversions can amplify gradient magnitudes substantially, foreshadowing the stability analysis of Chapter 4.

Chapter 4

Methods

This chapter details the methodology introduced to integrate hyperbolic geometry into the quantization process of Residual Vector Quantized Variational Autoencoders (RQ-VAEs). Building on the background of Chapter 3, we lift VQ and RVQ onto the Poincaré ball (§4.1) and then concentrate on the gradient that the quantizer must propagate to the encoder. We show that the Euclidean straight-through estimator is geometrically invalid on a curved manifold and replace it with a parallel-transport estimator that is additionally stabilized against the catastrophic cancellation it incurs near the boundary (§4.1.2). Making the estimator exact, however, dismantles the residual gradient *firewall* of §3.2; we analyze the resulting gradient leakage, its compounding across stages, and the correction that restores stability (§4.2). We then justify the Möbius residual convention assumed throughout (§4.3) and describe the hyperbolic training objectives that enforce hierarchical inductive biases (§??).

4.1 Hyperbolic Residual Quantization

Despite their success, standard VQ and RVQ operate in Euclidean space, which can lead to vanishing residuals and structural distortions when embedding hierarchical data. To address this, we extend the quantization operations to the Poincaré ball manifold $\mathbb{D}_c^D$, where $c > 0$ represents the absolute value of the negative curvature.

4.1.1 Hyperbolic Distance and Quantization

In Hyperbolic VQ, the Euclidean distance used to match the continuous vector to the codebook centroids is replaced by the squared hyperbolic distance metric. Given a vector $x \in \mathbb{D}_c^D$ and a codebook $C \subset \mathbb{D}_c^D$, the nearest neighbor assignment is computed as:

$$q(x) = c_k, \quad \text{where } k = \operatorname{argmin}_j d_{\mathbb{D}_c}^2(x, c_j) \quad (4.1)$$

This formulation ensures that the assignment process strictly follows the geodesics of the hyperbolic manifold.

Lifting the residual recursion of Eq. 3.4 to the manifold appears, at first sight, to be a matter of transliteration: replace Euclidean subtraction with Möbius subtraction and Euclidean summation with Möbius addition while preserving the order of operations. This is the convention adopted by prior hyperbolic residual quantizers such as HRQ [?]. Writing $r_0 = z_e$ for the (projected) encoder output, stage i selects $q_i = q(r_{i-1})$ and forms the next residual by *right* Möbius subtraction, while the output reaggregates the codewords by left-associated Möbius addition in coarse-to-fine order:

$$r_i = r_{i-1} \ominus_c q_i = r_{i-1} \oplus_c (-q_i), \quad \hat{z} = (\cdots (q_1 \oplus_c q_2) \oplus_c \cdots) \oplus_c q_N. \quad (4.2)$$

These replace the Euclidean $r_i = r_{i-1} - q_i$ and $\hat{z} = \sum_i q_i$ of §3.2 and keep every intermediate quantity on the ball; every Möbius output is clipped back into the open ball by a projection proj_c (a rescaling to radius $(1 - \epsilon)/\sqrt{c}$) to maintain numerical validity.

The residual mismatch. This transliteration is faithful in Euclidean space but breaks on the ball. The purpose of the residual is to track the portion of the encoder point not yet captured by the running reconstruction $\hat{z}_i = (\cdots (q_1 \oplus_c q_2) \cdots) \oplus_c q_i$ formed from the first i codes; the genuine remaining error is the *true residual*

$$R_i^{\text{true}} := (-\hat{z}_i) \oplus_c z_e, \quad \text{so that} \quad \hat{z}_i \oplus_c R_i^{\text{true}} = z_e. \quad (4.3)$$

In Euclidean RVQ the recursively tracked residual coincides with it, $r_i = z_e - \sum_{j \leq i} q_j = R_i^{\text{true}}$, because addition is commutative and associative; the decomposition then telescopes exactly and $\hat{z} + r_D = z_e$. On the Poincaré ball neither property holds: right Möbius subtraction is *not* inverted by left-associated Möbius addition, so the tracked residual r_i of Eq. 4.2 departs from the true residual R_i^{true} of Eq. 4.3 by a gyration that the convention never accounts for, $r_i \neq R_i^{\text{true}}$. This mismatch is not a transient numerical artefact: it is incurred at every stage and compounds with depth, so $\hat{z} \oplus_c r_D$ drifts away from z_e even when every codeword is quantized without error. As a consequence, each codebook beyond the first is fitted to quantize a residual r_i that no longer represents the actual reconstruction error R_i^{true} , and the deeper the stack the further the quantization target diverges from the quantity it is meant to encode. This silently erodes the coarse-to-fine hierarchy that motivates residual quantization in the first place. Resolving the mismatch—choosing the unique pairing of Möbius residual and aggregation for which $r_i = R_i^{\text{true}}$ and the cascade telescopes exactly—is the subject of §4.3.

4.1.2 The Hyperbolic Straight-Through Estimator

The nearest-neighbour assignment is non-differentiable, and in Euclidean VQ-VAE the gradient is delivered to the encoder by the straight-through estimator of Eq. 3.2, whose Jacobian is the identity. Transplanting this estimator verbatim to the Poincaré ball is geometrically invalid for two related reasons. First, the additive correction $z_e + \text{sg}[q(z_e) - z_e]$ is ordinary vector addition, which does not respect the manifold and can leave the ball. Second, and more fundamentally, the identity Jacobian implicitly identifies the tangent spaces $T_{z_e} \mathbb{D}_c^D$ and $T_q \mathbb{D}_c^D$ as the same vector space. They are not: a cotangent vector defined at the code q cannot be equated with one at z_e without parallel transport (§3.3), and the conformal factor differs between the base points, $\lambda_{z_e}^c \neq \lambda_q^c$. Copying the gradient unchanged therefore corrupts both its magnitude—it omits the λ -rescaling implied by the Riemannian metric (Eq. 3.14)—and its direction—it omits the gyration carried by transport (Eq. 3.13).

Geometrically valid forward pass. We keep the estimator an exact identity on the manifold by routing the stop-gradient through the tangent space at z_e rather than through Euclidean coordinates:

$$\hat{z}_{\text{STE}} = \exp_{z_e}^c \left(\text{sg} \left[\log_{z_e}^c q(z_e) \right] \right). \quad (4.4)$$

Since $\exp_{z_e}^c$ inverts $\log_{z_e}^c$ (§3.3), this evaluates exactly to $q(z_e)$ in the forward pass, while the stop-gradient tangent vector decouples the discrete choice from the graph.

Parallel-transport backward pass. The substance of the estimator is its backward pass. The decoder delivers a Euclidean gradient $g_q := \partial L / \partial q$ at the code q ; to move it to z_e correctly we apply the three-step Riemannian procedure of §3.3 (Eqs. 3.13 and 3.14): (i) convert g_q to a Riemannian gradient at q by dividing by $(\lambda_q^c)^2$; (ii) parallel-transport the resulting tangent

vector from q to z_e along the connecting geodesic, which rescales it by $\lambda_q^c/\lambda_{z_e}^c$ and rotates it by $\text{gyr}[z_e, -q]$; and (iii) convert back to a Euclidean gradient at z_e by multiplying by $(\lambda_{z_e}^c)^2$. Composing the three steps,

$$\frac{\partial L}{\partial z_e} = (\lambda_{z_e}^c)^2 P_{q \rightarrow z_e}^c \left(\frac{1}{(\lambda_q^c)^2} g_q \right) = \frac{\lambda_{z_e}^c}{\lambda_q^c} \text{gyr}[z_e, -q] g_q. \quad (4.5)$$

We call the estimator defined by Eqs. 4.4–4.5 the *hyperbolic straight-through estimator* (HSTE). The Euclidean STE is recovered only in the flat limit, where λ^c becomes constant and the gyration collapses to the identity, so that Eq. 4.5 reduces to $\partial L/\partial z_e = g_q$. A practical variant, used when the encoder is updated by a Riemannian optimizer, omits the re-conversion of step (iii) and returns the Riemannian gradient $P_{q \rightarrow z_e}^c(g_q/(\lambda_q^c)^2)$ directly; this cancels the $(\lambda_{z_e}^c)^2$ factor at the source and avoids the boundary amplification analyzed in §4.2.

Numerically stable gyration. The gyration in Eq. 4.5 is the numerically delicate component, precisely because q is a quantization of z_e and the two points are close. Writing the closed-form gyration as $\text{gyr}[A, B]v = v + 2(aA + bB)/d$ with $A = q$ and $B = -z_e$, the scalar coefficients and denominator are

$$\begin{aligned} a &= -c^2 \langle A, v \rangle \|B\|^2 + c \langle B, v \rangle + 2c^2 \langle A, B \rangle \langle B, v \rangle, \\ b &= -c^2 \langle B, v \rangle \|A\|^2 - c \langle A, v \rangle, \\ d &= 1 + 2c \langle A, B \rangle + c^2 \|A\|^2 \|B\|^2. \end{aligned} \quad (4.6)$$

Setting $\delta := q - z_e$ (the quantization error, with $\|\delta\|$ small) gives $A \approx -B$, which triggers catastrophic cancellation twice. The denominator collapses to $d \approx (1 - c\|z_e\|^2)^2$, evaluated as a difference of $O(1)$ terms that itself vanishes as embeddings approach the boundary ($1 - c\|z_e\|^2 \rightarrow 0$); and the numerator $aA + bB = aq - bz_e$ subtracts two nearly identical large vectors, since $a \approx b$. Dividing the resulting floating-point noise by the ill-conditioned denominator makes the transported gradient explode. We restore precision by re-expressing both quantities exactly in terms of δ . Cancelling the $O(1)$ terms analytically yields the stable denominator

$$d = (1 - c\|z_e\|^2)^2 - 2c(1 - c\|z_e\|^2) \langle z_e, \delta \rangle + c^2 \|z_e\|^2 \|\delta\|^2, \quad (4.7)$$

a sum of small terms of comparable magnitude, and, using $aA + bB = \frac{a+b}{2} \delta + \frac{a-b}{2} (q + z_e)$,

$$a - b = c(1 - c\|z_e\|^2) \langle \delta, v \rangle - c^2 \langle z_e, v \rangle \|\delta\|^2, \quad (4.8)$$

in which the offending $O(1)$ contributions cancel before evaluation rather than after; the combination $a + b$ is computed directly, as a and b share a sign. Equations 4.7–4.8 leave the estimator mathematically identical to Eq. 4.5 but keep it finite at the boundary, where the model is most expressive. This stabilization is necessary but, as the next section shows, not sufficient: it controls the error of a *single* transport, while a stacked quantizer composes N of them.

4.2 The Residual Firewall and Gradient Routing

Replacing the Euclidean STE with the geometrically exact HSTE removes one approximation but exposes a deeper structural problem in how gradients are routed through a residual stack. This section analyzes that problem and the correction (`gradient_correction`) that resolves it.

4.2.1 The Euclidean firewall and its breakdown

Recall from §3.2 the residual gradient *firewall* of Euclidean RVQ: combining the identity STE Jacobian with $r_i = r_{i-1} - q_i$ gives $\partial r_i / \partial r_{i-1} = I - I = 0$ (Eq. 3.5), so the residual branch transmits no gradient and the encoder receives exactly one clean copy of the reconstruction signal regardless of depth. We now show this cancellation is an artefact of flat geometry. Under the HSTE the quantizer Jacobian is the parallel transport $\partial q_i / \partial r_{i-1} = P_{q_i \rightarrow r_{i-1}}^c$, and the residual is the Möbius update $r_i = (-q_i) \oplus_c r_{i-1}$ (Eq. 4.14). Differentiating through both arguments of $\oplus_c$,

$$A_i := \frac{\partial r_i}{\partial r_{i-1}} = \underbrace{D_2 \oplus_c (-q_i, r_{i-1})}_{\text{direct path}} - \underbrace{D_1 \oplus_c (-q_i, r_{i-1}) \circ P_{q_i \rightarrow r_{i-1}}^c}_{\text{path through } q_i}, \quad (4.9)$$

where $D_1 \oplus_c$ and $D_2 \oplus_c$ denote the Jacobians of Möbius addition in its first and second arguments. In Euclidean space $D_1 = D_2 = I$ and $P = I$, recovering $A_i = I - I = 0$. On the ball the two terms no longer match: the differentials carry distinct conformal factors ($\lambda_{r_{i-1}}^c \neq \lambda_{q_i}^c$ in general), and—decisively—the transport $P_{q_i \rightarrow r_{i-1}}^c$ applies a gyration that rotates the gradient, whereas the direct term does not. Subtracting a rotated vector from an unrotated one cannot vanish, so $A_i \neq 0$: the firewall leaks.

Why the Euclidean STE hid the leak. It is worth asking why this instability did not surface when the same hyperbolic model was trained with the ordinary Euclidean STE ($\partial q_i / \partial r_{i-1} = I$). Because $q_i \approx r_{i-1}$, the next residual lands near the origin, where the Möbius differentials simplify to $D_1 \oplus_c (z, -z) = D_2 \oplus_c (z, -z) = \frac{\lambda_z^c}{2} I$. With the un-rotated identity Jacobian the two terms in Eq. 4.9 reduce to $\frac{\lambda_z^c}{2} I - \frac{\lambda_z^c}{2} I = 0$: the Euclidean STE *accidentally* reproduces the firewall through a first-order cancellation near the origin. The exact HSTE forfeits this accident precisely because it is correct—its gyration rotates the gradient out of alignment, so the scalar factors can no longer cancel. The leak is thus the price of geometric fidelity, not a bug in the transport.

4.2.2 Compounding across stacked transports

A single non-zero A_i would be tolerable; the danger is that the leak is incurred once per stage and composed across the entire cascade. Unrolling $r_0 = z_e$, $q_i = \text{HSTE}(r_{i-1}, c_i)$, $r_i = (-q_i) \oplus_c r_{i-1}$ with output $\hat{z} = q_1 \oplus_c (\cdots \oplus_c q_N)$, the reconstruction gradient that reaches the encoder is

$$\nabla_{r_0} L_{\text{rec}} = \sum_{i=1}^N \left(\prod_{j=1}^{i-1} A_j \right)^\top (P_{q_i \rightarrow r_{i-1}}^c)^\top \nabla_{q_i} L_{\text{rec}}, \quad \prod_{j=1}^{i-1} A_j := A_{i-1} \cdots A_1, \quad (4.10)$$

with the empty product ($i = 1$) equal to I . The gradient of the i -th quantizer must traverse a chain of $i - 1$ leak Jacobians before reaching the encoder. Two limits bound the behaviour: with the firewall intact ($A_j = 0$) every term with $i \geq 2$ vanishes and the encoder receives the single clean gradient $(P_{q_1 \rightarrow z_e}^c)^\top \nabla_{q_1} L_{\text{rec}}$, exactly as in Euclidean RVQ; with $A_j \neq 0$ all N terms survive, each filtered through a different-length product of leak matrices.

4.2.3 Spectral radius and the conformal amplifier

Taking norms in Eq. 4.10 with $\rho := \max_j \|A_j\|$ and representative transport and gradient scales τ, g ,

$$\|\nabla_{r_0} L_{\text{rec}}\| \lesssim \tau g \sum_{i=1}^N \rho^{i-1} = \tau g \frac{\rho^N - 1}{\rho - 1}, \quad (4.11)$$

so depth N enters as an exponent and the spectral radius of the leak matrices decides the regime: for $\rho > 1$ the deep quantizers deliver exponentially amplified gradients (*explosion*); for $\rho < 1$ they are exponentially attenuated, so the encoder effectively sees only the first few stages (a *vanishing-horizon* effect, present even without a firewall); $\rho = 1$ is a non-generic knife edge. Crucially ρ is not $O(1)$. Tracing the HSTE backward (Eq. 4.5), the net transport scales with the conformal ratio $\lambda_{r_{i-1}}^c/\lambda_{q_i}^c$, and the Möbius differentials contribute further factors $\lambda_{r_{i-1}}^c/\lambda_0^c$. As any base point migrates toward the boundary, $\|z\|^2 \rightarrow 1/c$ and $\lambda_z^c \rightarrow \infty$, so both τ and ρ diverge. The cascade thus contains a built-in amplifier that is strongest exactly where the geometry is most expressive (near the boundary) and where the quantization error is largest (early training)—the same regime that made the bare gyration of §4.1.2 numerically unstable. The two failure modes are moreover multiplicative: writing the realized backward map as $\hat{P} = P + E$ for the residual clamping and finite-precision error E , the stacked product $\prod_j (A_j + E_j)$ amplifies E by the very products $\prod_j A_j$ that already drive the explosion. Improving the exactness of a single transport—as the stable gyration of §4.1.2 does—therefore cannot tame a product whose spectral radius exceeds one.

4.2.4 Restoring the firewall: the gradient-routing correction

The analysis isolates a single decisive quantity—the spectral radius of the leak matrices A_j —and a single structural cause: the non-zero $\partial r_i/\partial r_{i-1}$ that the curved geometry produces. We remove it directly. By applying the stop-gradient to the residual after each stage, $r_i \leftarrow \text{sg}[r_i]$ (the `gradient_correction` option), we force $A_j := 0$ by construction, truncating the sum in Eq. 4.10 to its $i = 1$ term:

$$\nabla_{z_e} L_{\text{rec}} = \left(P_{q_1 \rightarrow z_e}^c \right)^\top \nabla_{q_1} L_{\text{rec}}. \quad (4.12)$$

This is the exact hyperbolic analogue of the Euclidean firewall (Eq. 3.5): the encoder again receives a single, holistic Riemannian gradient—holistic because $\hat{z}$, and hence $\nabla_{q_1} L_{\text{rec}}$, already depends on every deeper code—transported once from q_1 . The stop-gradient does not starve the deeper codebooks: each q_i still receives its reconstruction gradient directly through $\partial \hat{z}/\partial q_i$, and the per-stage codebook and commitment losses (§??) attach to q_i and r_{i-1} with their own stop-gradients, so the deep dictionaries continue to train while their gradients stop at the firewall instead of leaking back to the encoder. Setting $A_j = 0$ simultaneously removes all three failure modes identified above: the spectral growth, the boundary conformal amplification, and the multiplicative compounding of transport error.

Applicability to standard VQ-VAEs. The correction is geometry-agnostic. In Euclidean RVQ the firewall holds automatically only because the STE Jacobian and the residual Jacobian are both the identity; the moment that exact $I - I$ cancellation is perturbed—by a non-identity straight-through map, a normalized or otherwise transformed residual, or a learned projection between stages—the same leakage of Eq. 4.10 reappears, now with a Euclidean rather than gyrational mismatch. Applying the stop-gradient to the residual restores the clean single-gradient routing in those settings exactly as it does here, at no cost in the unperturbed case, where it is a no-op in exact arithmetic. The gradient-routing correction is thus a drop-in stabilizer for any residual quantizer, hyperbolic or not, that makes the encoder’s gradient exact by construction rather than reliant on an accidental cancellation.

4.3 Residual Aggregation: the `new_method` Convention

It remains to justify the specific ordering of the Möbius residual and aggregation that the `new_method` option selects, and that the firewall analysis of §4.2 already assumed (Eq. 4.14

below). In Euclidean RVQ the residual update $r_i = r_{i-1} - q_i$ and the sum $\hat{z} = \sum_i q_i$ are mutually inverse for free: addition is commutative and associative, so the order in which codewords are removed and re-added is irrelevant and $\hat{z} + r_N = z_e$ holds exactly. On the Poincaré ball neither property holds, and the naive transliteration of §4.1 (Eq. 4.2) breaks the telescoping that makes residual quantization a faithful coarse-to-fine decomposition, so that the tracked residual no longer equals the true residual R_i^{true} of Eq. 4.3.

The requirement is that aggregation invert the residual recursion, so that reconstructing from the codes returns the encoder point up to the final residual. The gyrogroup left-cancellation law,

$$a \oplus_c ((-a) \oplus_c b) = b, \quad (4.13)$$

makes the correct pairing explicit. Defining the residual by *left* Möbius subtraction and reconstructing by reverse-nested aggregation,

$$r_i = (-q_i) \oplus_c r_{i-1}, \quad \hat{z} = q_1 \oplus_c (q_2 \oplus_c (\cdots \oplus_c q_N)), \quad (4.14)$$

restores the exact inverse pair that the previous-work convention of Eq. 4.2 lacks. The first stage gives $r_1 = (-q_1) \oplus_c z_e$, which Eq. 4.13 inverts as $q_1 \oplus_c r_1 = z_e$. Applying this telescopically, the full reconstruction $q_1 \oplus_c (q_2 \oplus_c (\cdots \oplus_c (q_N \oplus_c r_N)))$ recovers z_e exactly; dropping the unquantized tail r_N yields the reconstruction $\hat{z}$ above. With this pairing the tracked residual equals the true residual at every stage, $r_i = R_i^{\text{true}}$ (Eq. 4.3), so the mismatch identified in §4.1 vanishes. The ordering is therefore not a free choice: left subtraction must be paired with this particular reverse-nested aggregation for the codes to compose back into the encoder point.

The alternative conventions fail for the same reason that would make them equivalent in Euclidean space. Right subtraction $r_i = r_{i-1} \oplus_c (-q_i)$ is *not* inverted by left re-addition, because $\oplus_c$ is non-commutative; and aggregating the codes in forward (coarse-to-fine) rather than reverse order misplaces the gyration factors introduced by non-associativity. In both cases the reconstructed $\hat{z}$ drifts from z_e by a gyration-dependent residual that grows with depth, degrading reconstruction and, through Eq. 4.9, perturbing the very Jacobians whose spectral radius governs the firewall. The `new_method` ordering is the choice that keeps the hyperbolic residual cascade an exact inverse pair, and is assumed throughout the remainder of this thesis.

Chapter 5

Experiments

Chapter 6

Conclusions

Appendix A

Optional appendix